Phase 1: Project Identification and Description

CIS 566 Software Architecture and Design Patterns

Phase 1 establishes a clear, feasible, and informed direction for the semester project. Teams may select an instructor-proposed project idea or propose an original topic.

The project does not have to follow one fixed format, but software architecture or design patterns must shape the central problem, method, artifact, or evaluation.

Project scope

Projects may focus on software design and implementation, architecture analysis and redesign, design-pattern application, tool or prototype development, comparative evaluation, literature review or an optional research-oriented study. The proposal must show a substantive connection to course concepts and lead to evidence that the team can evaluate.

students who select a research-oriented project may have the opportunity to develop their work into a paper or another scholarly output after the course project is completed.

Requirements for every project

- Architecture or design patterns are central to the project rather than added only to satisfy the course requirement.
- The project produces a concrete contribution, such as a design, implementation, dataset, taxonomy, tool, analysis, validated method, or set of empirical findings.
- The scope is feasible within the semester and distinct from examples and prior course projects

Acceptable project pathways

Pathway	Possible focus	Expected architecture or pattern connection
Design and implementation	Build or extend a software system or prototype.	Compare architectural alternatives, justify selected patterns, and evaluate resulting quality attributes.
Architecture analysis and refactoring	Study an existing system, repository, or architecture.	Identify patterns, antipatterns, smells, erosion, or technical debt and validate a proposed redesign.
Empirical research	Conduct a mapping review, repository study, case study, survey, or experiment.	Investigate a focused question about architectural decisions, pattern use, consequences, or evolution.
Tool or method	Create a detector, recommender, visualization, benchmark, or analysis method.	Recover, detect, recommend, explain, or evaluate architecture and patterns.
Student defined hybrid	Combine research and development in another defensible form.	Explain the course connection clearly and obtain instructor approval before proceeding.

Phase 1 report content

Use the following numbered sections in the report. Write specific statements that another reader can evaluate. Early decisions may change in later phases, but the Phase 1 plan should be coherent and supported by preliminary evidence.

1. Project identity and pathway

Provide the project title, team members, selected pathway, and a two to four sentence project summary.

- If the topic comes from an instructor pitch, identify the pitch and explain the team specific direction.

2. Problem and motivation

Describe the software problem, who or what it affects, and why the problem deserves study.

3. Description of the proposed idea

4. Outcomes and deliverables

Explain what the team expects to contribute and list the artifacts that will support that contribution.

5. References

Provide complete and consistent citations for all sources used in the proposal.